Giri's Tech Hub, Pune.

Core Java Machine Test

Batch: Jan/Feb/March-26

Date : 17/08/2026
Time : 08 to 11.00 Am.

Instructions:

1. Solve any 9 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

Q1. Write a java program to check number is Armstrong number or not happy using function recursion.

Q2. Write a java program to print this pattern.

```
        1
      2   2
    3       3
  4           4
5               5
  4           4
    3       3
      2   2
        1
```

Q3. Given an integer array containing positive and negative numbers, find the contiguous subarray having the maximum sum and display the subarray and its sum.
Explanation:
Traverse the array while maintaining the current sum and maximum sum. Reset the current subarray when continuing it would reduce the possible maximum.
Input:
int[] arr = {-2, 1, -3, 4, -1, 2, 1, -5, 4};
Output:
Maximum Sum Subarray = {4, -1, 2, 1}
Maximum Sum = 6

Q4. Given an array of strings, find the string containing the maximum number of unique characters. Ignore character case while checking uniqueness.
Explanation:
For every string, count its unique characters. Compare the count with the previous maximum and return the string having the highest unique-character count.
Input:
{"apple", "developer", "banana", "python", "programming"}
Output:
String = developer
Unique Characters = 7

Q5. Create an Employee POJO class with id, name, department, salary, and experience. Implement CRUD operations using class , object , pojo class. Add a search operation to find employees based on department and minimum salary.

Explanation:

Create an Employee class with private fields, constructors, getters and setters. Maintain employee objects inside an ArrayList. Implement:

- **Add Employee**
- **Update Employee by ID**
- **Delete Employee by ID**
- **Display all employees**
- **Search employees by department**

Q6. Create an abstract class Employee with employee details and an abstract method calculateSalary(). Create an interface Bonus with method calculateBonus(). Implement both in PermanentEmployee and ContractEmployee.

Explanation:

- **Permanent employees receive a bonus of 15% of salary.**
- **Contract employees receive a bonus of 5%.**
- **Use constructor to initialize employee information.**
- **Calculate salary + bonus.**
- **Display employee type, salary, bonus and final amount.**

Input:

Permanent Employee:

Name = Rahul

Salary = 60000

Contract Employee:

Name = Amit

Salary = 50000

Output:

Permanent Employee:

Name = Rahul

Salary = 60000

Bonus = 9000

Final Salary = 69000

Contract Employee:

Name = Amit

Salary = 50000

Bonus = 2500

Final Salary = 52500

Q7. Given an ArrayList<String>, find the longest string and shortest string without using Collections.sort() or any built-in sorting method. If multiple strings have the same length, display all of them.

Explanation:

Maintain maxLength and minLength while traversing the list. Store all strings matching the maximum and minimum lengths.

Input: ["Java", "Python", "SQL", "Programming",

"AI", "Developer", "Data", "Analytics"]

Output:

Longest String: Programming

Shortest Strings: AI

Q8. Given an ArrayList<String>, reverse every individual string using only while loops. Do not use StringBuilder.reverse() or any built-in reverse method.
Explanation:
Traverse the ArrayList using one while loop. For each string, start from its last character and move toward the first character using another while loop.
Input:
["Java", "Python", "SQL", "Developer", "Data"]
Output:
[avaJ, nohtyP, LQS, repoleveD, ataD]

Q9. Store N Words in a HashMap. Print Words whose length is prime number.
Input :java python sql codex spring
Output: Words with Prime Length
sql-3
codex-5

Q10. Create a Map<Integer, Teacher> where the Teacher ID is the key and the Teacher object is the value. The Teacher class should contain Teacher Name and Salary.
Perform the following operations:

1. **Create a Map<Integer, Teacher> and add multiple teacher records.**
2. **Convert the Map values into a List<Teacher>.**
3. **Sort the list of teachers based on Salary in ascending order using Comparable or Comparator.**
4. **Display the sorted list of teachers.**

-----ALL THE BEST-----